

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1472

Huntley 138 kV
January 29, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1472. The Interconnection Customer proposes a 181.9 MW solar photovoltaic facility interconnecting at the Huntley 138 kV substation in the Virginia area. The request was received on May 05, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 431 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,368,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Old Chapel to Hp Hood Ct 138.0 kV Line
Contingency: Twin Lakes Alte to Huntley 138.0 kV
Loading: 105.7% (Rating: 291 MVA)
DFAX: 9.4%
Pre-project loading: 99.9%
2. Facility: Ocean Rod to Monolith 138.0 kV
Contingency: International Paper Valliant Ok to Kearneyville 345.0 kV
Loading: 101.7% (Rating: 314 MVA)
DFAX: 5.2%
Pre-project loading: 98.7%
3. Facility: Pasco Cnty Solid Waste Resource Recovery to Novec Paradise 138.0 kV
Contingency: Pasco Cnty Solid Waste Resource Recovery to Vmea Peaking Gen 138.0 kV
Loading: 101.7% (Rating: 335 MVA)
DFAX: 16.2%
Pre-project loading: 92.9%
4. Facility: Ocean Rd to Three Peaks 138.0 kV
Contingency: Kearneyville to International Paper Valliant Ok 345.0 kV
Loading: 101.7% (Rating: 297 MVA)
DFAX: 5.2%
Pre-project loading: 98.5%
5. Facility: University Of Maine to Three Peaks 138.0 kV Line
Contingency: International Paper Valliant Ok to Kearneyville Substation 345.0 kV line
Loading: 101.6% (Rating: 293 MVA)
DFAX: 5.2%
Pre-project loading: 98.4%

6. Facility: Plum Pt Energy Sta to University Of Maine 138.0 kV Line
Contingency: Kearneyville to International Paper Valliant Ok Substation 345.0 kV
Loading: 101.5% (Rating: 286 MVA)
DFAX: 5.2%
Pre-project loading: 98.2%
7. Facility: Mabton to Plum Pt Energy Station 138.0 kV
Contingency: International Paper Valliant Ok to Kearneyville Substation 345.0 kV
Loading: 101.5% (Rating: 270 MVA)
DFAX: 5.2%
Pre-project loading: 98.0%
8. Facility: Twin Lakes Alte to T.S. Power 138.0 kV Line
Contingency: Loudn to Thomaston 345.0 kV L
Loading: 101.5% (Rating: 401 MVA)
DFAX: 34.6%
Pre-project loading: 85.9%
9. Facility: Mammoth Pacific 2 to T.S. Power Substation 138.0 kV Line
Contingency: Loudon to Thomaston 345.0 kV
Loading: 101.5% (Rating: 379 MVA)
DFAX: 34.6%
Pre-project loading: 84.9%
10. Facility: Wheeler to Mammoth Pacific 2 138.0 kV Line
Contingency: Loudon to Thomaston 345.0 kV
Loading: 101.5% (Rating: 349 MVA)
DFAX: 34.6%
Pre-project loading: 83.5%
11. Facility: Mabton to Southampton 138.0 kV
Contingency: International Paper Valliant Ok to Kearneyville 345.0 kV
Loading: 101.4% (Rating: 262 MVA)
DFAX: 5.2%
Pre-project loading: 97.8%
12. Facility: Celeron to Southmpton 138.0 kV
Contingency: International Paper Valliant Ok to Kearneyville Substation 345.0 kV Line
Loading: 101.3% (Rating: 252 MVA)
DFAX: 5.2%
Pre-project loading: 97.5%
13. Facility: Creston Nelson to Wpcp 138.0 kV
Contingency: Santa Rosa Energy Center to Balch 138.0 kV Line
Loading: 100.9% (Rating: 285 MVA)
DFAX: 5.0%
Pre-project loading: 97.7%
14. Facility: Creston Nelson to 3 Mile Knoll Substation 138.0 kV
Contingency: Santa Rosa Energy Ctr to Balch 138.0 kV
Loading: 100.8% (Rating: 275 MVA)
DFAX: 5.0%
Pre-project loading: 97.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hp Hood Ct to Old Chapel 138.0 kV	Reconductor line	\$4,762,000	100% (\$4,762,000)
Ocean Rod to Monolith 138.0 kV	Reconductor line	\$487,000	19% (\$94,000)
Pasco Cnty Solid Waste Resource Recovery to Novec Paadise 138.0 kV	Reconductor line	\$1,163,000	19% (\$220,000)
Ocean Road to Three Peaks 138.0 kV	Reconductor line	\$1,526,000	19% (\$295,000)
University Of Maine to Three Peaks 138.0 kV Line	Reconductor line	\$1,447,000	19% (\$279,000)
Plum Point Energy Station to UniversityOf Maine Substation 138.0 kV	Reconductor line	\$384,000	19% (\$74,000)
Mabton to Plum Poin Energy Sta 138.0 kV	Reconductor line	\$1,006,000	19% (\$194,000)
Twin Lakes Alte to T.S. Power Substation 138.0 kV	Reconductor line	\$1,455,000	17% (\$249,000)
Mammoth Pacific 2 to T.S. Power Substation 138.0 kV Line	Reconductor line	\$1,187,000	17% (\$203,000)
Wheeler to Mmmoth Pacific 2 138.0 kV	Reconductor line	\$3,214,000	17% (\$549,000)
Mabton to Southampton 138.0 kV	Reconductor line	\$1,242,000	19% (\$240,000)
Southampton to Celeron 138.0 kV L	Reconductor line	\$1,083,000	19% (\$209,000)
Creston Nelson to Wpcp 138.0 kV	Reconductor line	\$1,567,000	54% (\$850,000)
Creston Nelson to 3 Mile Knoll 138.0 kV	Reconductor line	\$2,121,000	54% (\$1,150,000)
	Total allocated to project		\$9,368,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.